of this measure. This measure is invariant to orthogonal transformations. Equivalence is indicated by a value of zero, and larger values indicate dissimilarity.

Relational Knowledge Loss. A common approach to transfer knowledge in the context of knowledge distillation is to train the student model to mimic the relations between the teacher's instance representations [49]. This is done by minimizing the total element-wise difference of RSMs with respect to a loss function $l : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}_+$:

$$m_{\text{RK}}(\boldsymbol{R}, \boldsymbol{R}') = \sum_{i,j=1}^{N} l\big(\boldsymbol{S}_{i,j}, \boldsymbol{S}'_{i,j}\big). \tag{34}$$

While we defined RSMs via pairwise similarities, this approach has notably been generalized to higher-dimensional RSMs. For example, Park et al. [112] considered three-dimensional RSMs $\boldsymbol{S} \in \mathbb{R}^{N \times N \times N}$, where each entry $\boldsymbol{S}_{i,j,k}$ corresponds to the cosine of the angle enclosed by the vectors $v_{i,j} = \boldsymbol{R}_i - \boldsymbol{R}_j$ and $v_{k,j} = \boldsymbol{R}_k - \boldsymbol{R}_j$, i.e., $\boldsymbol{S}_{i,j,k} = \text{cos-sim}(v_{i,j}, v_{j,k})$. Similarities are then aggregated over all instance triples. This measure instantiation is invariant to orthogonal transformations, translations, and scaling. Equivalence is indicated by a value of zero, larger values indicate dissimilarity.

3.4 Neighborhood-Based Measures

The measures in this section compare the nearest neighbors of instances in the representation space. More precisely, each of these measures determine the k nearest neighbors of each instance representation $\boldsymbol{R}_i$ in the full representation matrix $\boldsymbol{R}$ with respect to a given similarity function s. In that context, the neighborhood size k is a parameter that has to be chosen for the application at hand. Letting $\boldsymbol{S}$ denote the RSM of representation $\boldsymbol{R}$, and w.l.o.g. assuming that higher values indicate more similar representations, we formally define the set of the k nearest neighbors of the instance representation $\boldsymbol{R}_i$ as the set $\mathcal{N}^k_{\boldsymbol{R}}(i) \subset \{j : 1 \leq j \leq N, j \neq i\}$ with $|\mathcal{N}^k_{\boldsymbol{R}}(i)| = k$ for which it holds that $\boldsymbol{S}_{i,j} > \boldsymbol{S}_{i,l}$ for all $j \in \mathcal{N}^k_{\boldsymbol{R}}(i), l \notin \mathcal{N}^k_{\boldsymbol{R}}(i) \cup \{i\}$. Once the nearest neighbors sets are determined, they are either compared directly, or one further considers distances of the representation $\boldsymbol{R}_i$ to its nearest neighbors. For each of these measures, we then obtain a vector of instance-wise neighborhood similarities $\big(v^k_{\text{NN-sim}}(\boldsymbol{R}, \boldsymbol{R}')_i\big)_{i \in \{1,\ldots,N\}}$, which are averaged over all instances to obtain similarity measures for the full representations $\boldsymbol{R}, \boldsymbol{R}'$:

$$m^k_{\text{NN-sim}}(\boldsymbol{R}, \boldsymbol{R}') = \frac{1}{N} \sum_{i=1}^{N} v^k_{\text{NN-sim}}(\boldsymbol{R}, \boldsymbol{R}')_i. \tag{35}$$

However, the instance-wise similarities and their distribution could also be inspected more closely to obtain additional insights [71]. For brevity, in all the measures that we introduce in the following, we only give a description of how the instance-wise similarities are computed. Similar to RSM-based measures, the choice of the similarity function s determines which transformations these measures are invariant to. By default, and in line with the literature, we assume the use of cosine similarity (Eq. A.5), which leads to invariance to orthogonal transformations and isotropic scaling.

k-NN Jaccard Similarity. This measure, also named *Nearest Neighbor Graph Similarity* [52] and *Nearest Neighbor Topological Similarity* [61], considers how many of the k nearest neighbors each instance has in common over a given pair of representations. The instance-wise neighborhood similarities are computed in terms of the Jaccard similarities of the neighborhood sets $\mathcal{N}^k_{\boldsymbol{R}}(i), \mathcal{N}^k_{\boldsymbol{R}'}(i)$:

$$\big(\boldsymbol{v}^k_{\text{Jac}}(\boldsymbol{R}, \boldsymbol{R}')\big)_i := \frac{|\mathcal{N}^k_{\boldsymbol{R}}(i) \cap \mathcal{N}^k_{\boldsymbol{R}'}(i)|}{|\mathcal{N}^k_{\boldsymbol{R}}(i) \cup \mathcal{N}^k_{\boldsymbol{R}'}(i)|}. \tag{36}$$

Jaccard similarity is bounded in the interval $[0, 1]$, with a value of one indicating identical neighborhoods. Aside from the commonly used cosine similarity [124, 149], Euclidean distance was also used as similarity function [61].

Second-Order Cosine Similarity. This measure was proposed by Hamilton et al. [53] to analyze changes in word embeddings over time. For each instance i, it first computes the union of nearest neighbors as an ordered set $\{j_1, \ldots, j_{K(i)}\} := \mathcal{N}^k_{\boldsymbol{R}}(i) \cup \mathcal{N}^k_{\boldsymbol{R}'}(i)$ in $\boldsymbol{R}$ and $\boldsymbol{R}'$ in terms of cosine similarity (Eq. A.5). Then the cosine similarities to these neighbors are compared between the two representations. Utilizing the cosine similarity RSMs $\boldsymbol{S}, \boldsymbol{S}'$ of the representations $\boldsymbol{R}, \boldsymbol{R}'$, the instance-wise second-order cosine similarities can then be defined as follows:

$$\big(\boldsymbol{v}^k_{\text{2nd-cos}}(\boldsymbol{R}, \boldsymbol{R}')\big)_i := \quad \text{cos-sim}\left(\big(\boldsymbol{S}_{i,j_1}, \ldots, \boldsymbol{S}_{i,j_{K(i)}}\big), \big(\boldsymbol{S}'_{i,j_1}, \ldots, \boldsymbol{S}'_{i,j_{K(i)}}\big)\right).$$

This measure is bounded in the interval $[0, 1]$, with $m^k_{\text{2nd-cos}}(\boldsymbol{R}, \boldsymbol{R}') = 1$ indicating equivalence of $\boldsymbol{R}$ and $\boldsymbol{R}'$.

Rather than considering the union of the neighborhood sets, Chen et al. [24] considered the intersection of the top-k neighborhoods. Another similar approach was presented by Moschella et al. [102], who used a random fixed set of reference instances instead of neighbors. Further, *Pointwise Normalized Kernel Alignment* (PNKA) [71] can be seen as a variant of second-order cosine similarity with $k = N$, but different similarity function s for the RSM.

Rank Similarity. The k-NN Jaccard similarity captures the extent to which two neighborhood sets overlap, but not the order of the common neighbors within those sets. To increase the importance of close neighbors, Wang et al. [149]

determined distance-based ranks $r_{\boldsymbol{R}_i}(j)$ to all $j \in \mathcal{N}^k_{\boldsymbol{R}}(i)$, where $r_{\boldsymbol{R}_i}(j) = n$ if $\boldsymbol{R}_j$ is the n-th closest neighbor of $\boldsymbol{R}_i$ with respect to a given similarity function s. Based on these ranks, they defined the instance-based similarities as

$$\left(\boldsymbol{v}^k_{\text{ranksim}}(\boldsymbol{R}, \boldsymbol{R}')\right)_i = \frac{1}{(\boldsymbol{v}_{\max})_i} \cdot \sum_{j \in \mathcal{N}^k_{\boldsymbol{R}}(i) \cap \mathcal{N}^k_{\boldsymbol{R}'}(i)} \frac{2}{(1+|r_{\boldsymbol{R}_i}(j) - r_{\boldsymbol{R}'_i}(j)|)(r_{\boldsymbol{R}_i}(j) + r_{\boldsymbol{R}'_i}(j))}, \tag{37}$$

where $(\boldsymbol{v}_{\max})_i = \sum_{k=1}^{K} \frac{1}{k}$, with $K = |\mathcal{N}^k_{\boldsymbol{R}}(i) \cap \mathcal{N}^k_{\boldsymbol{R}'}(i)|$, is a normalization factor that limits the maximum of the ranking similarity to one, which is achieved for completely identical rankings. Intuitively, the first factor of the denominator in Equation (37) measures the similarity of the ranks of an instance, whereas the second factor assigns rank-based weights to this similarity, with lower-ranked instances gaining less influence.

Joint Rank and k-NN Jaccard Similarity. Rank similarity has the issue that it is only calculated on the intersection of the k-nearest neighbor sets in different representations. That means rank similarity might be high, even if the k-NN sets have almost no overlap. Similarly, Jaccard similarity might be high, but the order of the nearest neighbors might be completely different. Therefore, Wang et al. [149] combined these two approaches to calculate the *Embedding Stability*, by considering the product of Jaccard and rank similarity. Thus, using the instance vectors defined in Equation (36) and Equation (37), we can define the vector of instance-wise similarities as

$$\left(\boldsymbol{v}^k_{\text{Jac-Rank}}(\boldsymbol{R}, \boldsymbol{R}')\right)_i = \left(\boldsymbol{v}^k_{\text{Jac}}(\boldsymbol{R}, \boldsymbol{R}')\right)_i \cdot \left(\boldsymbol{v}^k_{\text{ranksim}}(\boldsymbol{R}, \boldsymbol{R}')\right)_i. \tag{38}$$

Scores are bounded in the interval $[0, 1]$, with $m^k_{\text{Jac-Rank}}(\boldsymbol{R}, \boldsymbol{R}') = 1$ indicating perfect similarity.

3.5 Topology-Based Measures

The measures in this category are motivated by the *manifold hypothesis* [48, Sec. 5.11.3], which states that high-dimensional representations are expected to be concentrated in the vicinity of a low-dimensional data manifold $\mathcal{M}$. Following this assumption, these measures then aim to approximate the manifolds in terms of discrete topological structures such as graphs, or, more generally, (abstract) simplicial complexes [56], based on which the representations can then be compared. A simplicial complex can be seen as a generalization of graphs, in which vertices may not only be paired by edges, but can also form higher-dimensional simplices. In both cases, each instance i typically corresponds to a vertex $v_i \in \mathcal{V}$, and edges/simplices are formed from instances that are close together in the representation space.

Geometry Score. The *Geometry Score* (GS) [66] characterizes representations by the number of *one-dimensional holes* in their data manifolds. To obtain this number of holes, the manifold is approximated in terms of simplicial complexes $\mathcal{S}_\alpha$, $\alpha > 0$, in which vertices v_i form a simplex, if the α-neighborhoods of their representations $\boldsymbol{R}_i$ overlap with each other. On this simplicial complex, the number of one-dimensional holes corresponds to the number of specific cycles in the complex and can be efficiently computed as the rank of its first homology group H_1.

Given that the number of holes may differ dependent on α, and that there is no ground-truth regarding which value of α yields the most accurate approximation of the manifold, Khrulkov and Oseledets [66] suggest varying the value α between 0 and $\alpha_{\max} \propto \max_{i,j} \|\boldsymbol{R}_i - \boldsymbol{R}_j\|_2$. For each number of holes k, they collect the longest intervals (α_1, α_2), in which the number of holes is constant at k, into sets $\mathcal{B}_k$. Then, the *relative living time* of k holes defined as $\text{RLT}(k, \boldsymbol{R}) = \frac{1}{\alpha_{\max}} \sum_{(\alpha_1,\alpha_2) \in \mathcal{B}_k} (\alpha_2 - \alpha_1)$ can be considered as the probability that k holes exist in the manifold. Since building simplicial complexes from large data is computationally challenging, Khrulkov and Oseledets [66] suggested sampling numerous subsets $\mathcal{I}$ of $n < N$ instances to built multiple so-called witness complexes with much lower number of simplices. Finally, one then considers the mean relative living times (MRLT) resulting from these complexes:

$$m_{\text{GS}}(\boldsymbol{R}, \boldsymbol{R}') = \sum_{k=0}^{k_{\max}-1} (\text{MRLT}(k, \boldsymbol{R}) - \text{MRLT}(k, \boldsymbol{R}'))^2, \tag{39}$$

where $k_{\max}$ denotes the maximum number of holes that is considered. The authors suggested using $k_{\max} = 100$, aggregating the RLTs from 10,000 complexes of $n = 64$ vertices each, and setting $\alpha_{\max} = \frac{1}{128} / \frac{N}{5000} \cdot \max_{i,j \in \mathcal{I}} \|\boldsymbol{R}_i - \boldsymbol{R}_j\|_2$ for each sample $\mathcal{I}$. This measure is bounded in the interval $[0, k_{\max}]$, with $m_{\text{GS}}(\boldsymbol{R}, \boldsymbol{R}') = 0$ indicating equivalent representations. Further, it is invariant to orthogonal transformations, isotropic scaling, and translations.

Multi-Scale Intrinsic Distance. The *Multi-Scale Intrinsic Distance* (IMD) [144] applies k-NN graphs $\mathcal{G}(\boldsymbol{R})$ as a proxy to characterize and compare the manifold of the representations. Specifically, Tsitsulin et al. [144] utilize the *heat kernel trace* on $\mathcal{G}(\boldsymbol{R})$ to compare representations, which is defined as $\text{hkt}_{\mathcal{G}(\boldsymbol{R})}(t) = \sum_i e^{-t\lambda_i}$, with λ_i as the eigenvalues of the normalized graph Laplacian of $\mathcal{G}(\boldsymbol{R})$. Similarity between the manifolds is then computed as a lower bound of the Gromov-Wasserstein distance, which can be expressed in terms of the heat kernel trace:

$$m_{\text{IMD}}(\boldsymbol{R}, \boldsymbol{R}') = \sup_{t>0} e^{-2(t+t^{-1})} |\,\text{hkt}_{\mathcal{G}(\boldsymbol{R})}(t) - \text{hkt}_{\mathcal{G}(\boldsymbol{R}')}(t)|. \tag{40}$$

Practically, Tsitsulin et al. [144] approximate $\text{hkt}_{\mathcal{G}(\boldsymbol{R})}(t)$ using the *Stochastic Lanczos Quadrature* [145], and obtain the supremum by sampling t from a parameter grid. They built the graph using the $k = 5$ nearest neighbors with respect

to Euclidean distance. The IMD has no upper bound; the minimum, which indicates maximal similarity, is zero. It is invariant to orthogonal transformations, isotropic scaling and translations.

Representation Topology Divergence. Similar to the geometry score, *Representation Topology Divergence* (RTD) [8] also considers persistence intervals of topological features of representations. However, in this approach graphs are applied for simplicial approximation of the representations, and the number of their connected components are the topological feature of interest. Specifically, Barannikov et al. [8] first compute RSMs with Euclidean distance, and normalize these by the 90th percentile of their values. Then, for a given distance threshold $\alpha > 0$, they construct a graph $\mathcal{G}^\alpha(\boldsymbol{R})$ with its adjacency matrix $\boldsymbol{A}$ defined as $\boldsymbol{A}_{i,j} = \boldsymbol{S}_{i,j} \cdot \mathbb{1}\{\boldsymbol{S}_{i,j} < \alpha\}$, and a union graph $\mathcal{G}^\alpha(\boldsymbol{R}, \boldsymbol{R}')$ with its adjacency matrix $\boldsymbol{A}$ defined as $\boldsymbol{A}_{i,j} = \min(\boldsymbol{S}_{i,j}, \boldsymbol{S}'_{i,j}) \cdot \mathbb{1}\{\min(\boldsymbol{S}_{i,j}, \boldsymbol{S}'_{i,j}) < \alpha\}$. If $\mathcal{G}^\alpha(\boldsymbol{R})$ and $\mathcal{G}^\alpha(\boldsymbol{R}, \boldsymbol{R}')$ differ in the number of their connected components, this is considered a topological discrepancy. For each specific discrepancy that occurs for varying values of α, the longest corresponding interval (α_1, α_2), for which this discrepancy persists, is collected in a set $\mathcal{B}(\boldsymbol{R}, \boldsymbol{R}')$. The total length of these intervals, denoted as $b(\boldsymbol{R}, \boldsymbol{R}') = \sum_{(\alpha_1,\alpha_2)\in\mathcal{B}(\boldsymbol{R},\boldsymbol{R}')} \alpha_2 - \alpha_1$, then quantifies similarity between two representations. The final RTD measure is constructed by subsampling K subsets $\mathcal{I}^{(k)}$ of $n < N$ instances each, and collecting the values $b\big(\boldsymbol{R}^{(k)}, \boldsymbol{R}'^{(k)}\big)$ derived from the representations $\boldsymbol{R}^{(k)} = (\boldsymbol{R}_i)_{i\in\mathcal{I}^{(k)}} \in \mathbb{R}^{n\times D}$ to form a measure $RTD(\boldsymbol{R}, \boldsymbol{R}') = \frac{1}{K}\sum_{i=1}^{K} b\big(\boldsymbol{R}^{(k)}, \boldsymbol{R}'^{(k)}\big)$. Because RTD is asymmetric, the authors proposed to use

$$m_{\text{RTD}}(\boldsymbol{R}, \boldsymbol{R}') = \tfrac{1}{2}(RTD(\boldsymbol{R}, \boldsymbol{R}') + RTD(\boldsymbol{R}', \boldsymbol{R})). \tag{41}$$

For hyperparameters, they suggested using $K = 10$ subsets of $n = 500$ representations each as default values. An RTD of zero indicates equivalent representations, with higher values indicating less similarity. By construction of the RSMs, RTD is invariant to orthogonal transformations, isotropic scaling, and translations.

3.6 Descriptive Statistics

Measures of this category deviate from all previous measures in a way that they describe statistical properties of either (i) individual representations $\boldsymbol{R}$, or (ii) measures of variance in the instance representations $\boldsymbol{R}_i$ over sets $\mathcal{R}$ of more than two representations. In case of (i), the statistics can be directly compared over pairs or sets of representations. For case (ii), one could aggregate or analyze the distribution of the instance-wise variations. While there are numerous statistics that could be used to compare representations, in the following we specifically outline statistics that have already been used to characterize representations in existing literature.

Intrinsic Dimension. The *intrinsic dimension* of a representation $\boldsymbol{R}$ corresponds to the minimal number of variables that are necessary to describe its data points. It can be defined as the lowest value $M \in \mathbb{N}, M < N$, for which the representation $\boldsymbol{R}$ lies in a M-dimensional manifold of $\mathbb{R}^N$ [21]. This statistic has its roots in social sciences [127] and information theory [11], and has since been applied in countless other fields, resulting in different variants, and numerous methods to estimate its exact values—for more details, we point the interested reader to the survey by Camastra and Staiano [21]. In the context of neural network analysis, different variants of the intrinsic dimension have been used as a tool to analyze the amount of information that is contained and processed within high-dimensional layers [88, 2, 9]. This statistic is invariant to affine transformations.

Magnitude. Wang et al. [149] characterized *magnitude* as the Euclidean length of instance representations $\boldsymbol{R}_i$. Consequently, they considered the length of the mean instance representation as a statistic for a representation $\boldsymbol{R}$:

$$m_{\text{Mag}}(\boldsymbol{R}) := \|\tfrac{1}{N}\textstyle\sum_{i=1}^{N} \boldsymbol{R}_i\|_2. \tag{42}$$

Aside from aggregating magnitude over all instances, they further proposed a measure to quantify the variance of the magnitude of instance-wise representations over multiple models. More precisely, given a set of representations $\mathcal{R}$, Wang et al. [149] measured the variance in the magnitudes of individual instances i as

$$m_{\text{Var-Mag}}(\mathcal{R}, i) = \tfrac{1}{\max_{\boldsymbol{R}\in\mathcal{R}} \|\boldsymbol{R}_i\|_2 - \min_{\boldsymbol{R}\in\mathcal{R}} \|\boldsymbol{R}_i\|_2} \cdot \sqrt{\tfrac{1}{|\mathcal{R}|}\textstyle\sum_{\boldsymbol{R}\in\mathcal{R}} (\|\boldsymbol{R}_i\|_2 - \overline{d_i}(\mathcal{R}))^2}, \tag{43}$$

where $\overline{d_i}(\mathcal{R}) = \frac{1}{|\mathcal{R}|}\sum_{\boldsymbol{R}\in\mathcal{R}} \|\boldsymbol{R}_i\|_2$ is the average magnitude of the representations of instance i in $\mathcal{R}$. As magnitude is unaffected by transformations that preserve vector length, this statistic is invariant to orthogonal transformations.

Concentricity. Wang et al. [149] proposed *concentricity* as a measure of the density of representations. It is based on measuring the cosine similarities of each instance representation $\boldsymbol{R}_i$ to the average representation, which we denote as $\alpha_i(\boldsymbol{R}) = \text{cos-sim}(\boldsymbol{R}_i, \frac{1}{N}\sum_{j=1}^{N} \boldsymbol{R}_j)$. Similar to magnitude, Wang et al. [149] then considered the mean concentricity

$$m_{\text{mConc}}(\boldsymbol{R}) := \tfrac{1}{N}\textstyle\sum_{i=1}^{N} \alpha_i(\boldsymbol{R}) \tag{44}$$

as a statistic for a single model, and measured the instance-wise variance of concentricity via

$$m_{\text{Var-Conc}}(\mathcal{R}, i) = \tfrac{1}{\max_{\boldsymbol{R}\in\mathcal{R}} \alpha_i(\boldsymbol{R}) - \min_{\boldsymbol{R}\in\mathcal{R}} \alpha_i(\boldsymbol{R})} \cdot \sqrt{\tfrac{1}{|\mathcal{R}|}\textstyle\sum_{\boldsymbol{R}\in\mathcal{R}} (\alpha_i(\boldsymbol{R}) - \bar{\alpha}_i(\mathcal{R}))^2}, \tag{45}$$